

Matrix functions and stability

Realizing arbitrary polynomial growth exponents by finite matrix families

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because finite families must realize every exponent at every length; community impact connects switched dynamics and asymptotic matrix-product growth.

Context and notation

The joint spectral radius of a nonempty compact set $M \subset \mathbb{C}^{d \times d}$ is

$$\hat{\rho}(M) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in M} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

For a compact nonempty $M \subset \mathbb{R}^{d \times d}$ with $\hat{\rho}(M) = 1$, define its maximal product norm at length k by

$$g_M(k) = \max_{A_1, \dots, A_k \in M} \|A_k \cdots A_1\|_2.$$

This problem concerns growth within one family over time. [MF-07](#) instead requests a dimension-dependent bound uniform across families.

Problem statement

For every real $\alpha \geq 0$, do there exist a positive integer d , a finite nonempty $M \subset \mathbb{R}^{d \times d}$ with $\hat{\rho}(M) = 1$, and constants $0 < c \leq C < \infty$ such that

$$ck^\alpha \leq g_M(k) \leq Ck^\alpha \quad \text{for every integer } k \geq 1?$$

Reference and status evidence

Varney and Morris, [On marginal growth rates of matrix products](#), §7, Question 2. Corollary 6.1 realizes exponent $1/3$; the all-exponents question remains posed. Searches for the title, authors, and marginal-growth exponents through 2026 located no complete answer. Infinite compact families and subsequence-only lower bounds do not meet the finite-family, every-length target.

Audit — 2026-09-10

Rechecked [Varney–Morris, Corollary 6.1, Proposition 3.1, and Question 2](#): exponent $1/3$ is realized and realizable exponents are closed under addition. The all-exponents assertion remains open. Author and marginal-growth-exponent searches found no complete finite-family construction or obstruction.